



MKS SERV0 42ES/57ES Closed-Loop Stepper Motor

Modbus RTU User Manual V1.0.1

Note: This manual corresponds to firmware version V 1.0.1 .

MKS SERV042ES /57ES RTU Version Notes			
Instruction manual version	content	Firmware version	date
V1.0.0	First release version	V1.0.0	Dec-2024
V1.0.1	1. Fixed some bugs	V1.0.1	Nov-2025
	2. Add IAP firmware upgrade function		
	3. F6 command speed mode adds the function of setting the runtime.		



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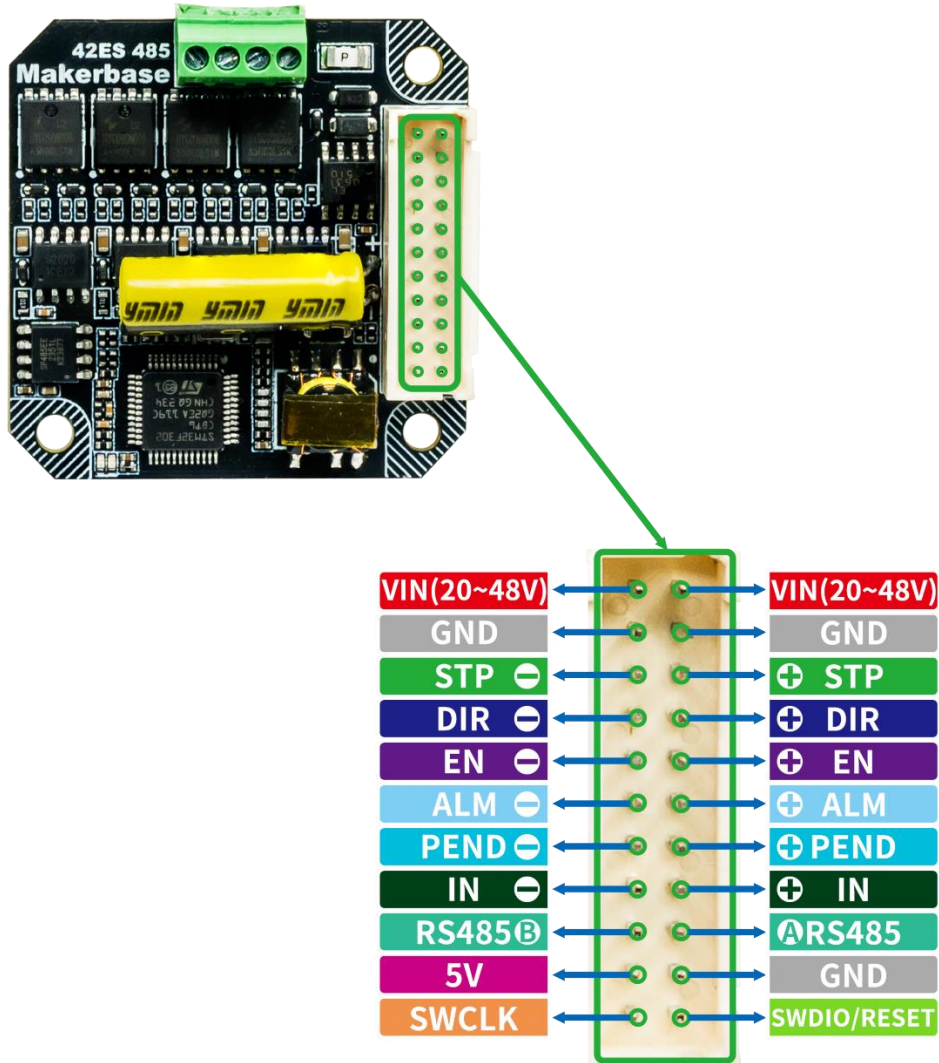


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Part 1. Port and wiring instructions

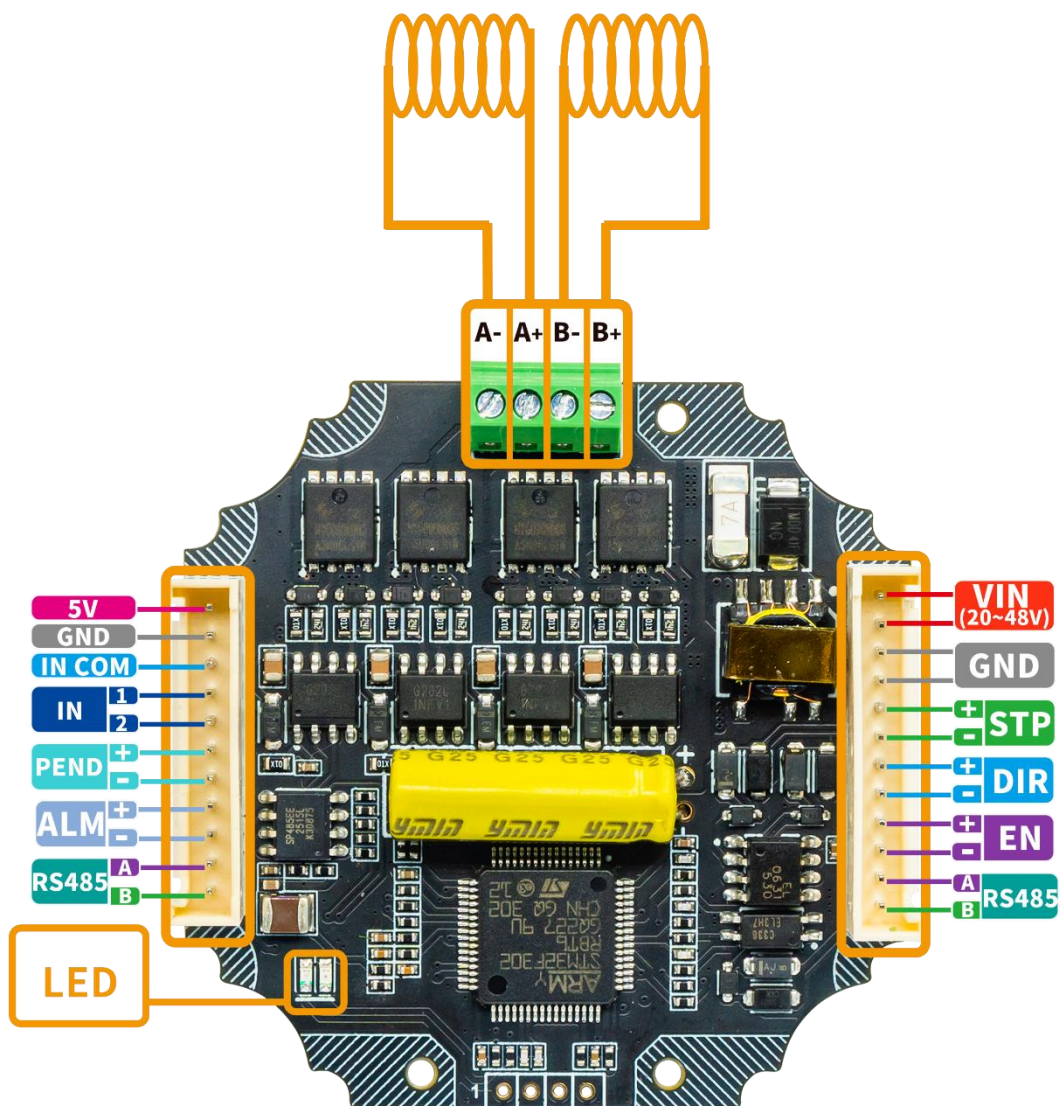
1.1 Interface Description

① SERV042ES_ RS485 Interface Description



1. Input power supply 20V~48V
2. 5V is the output power (5V, 100mA).
3. ALM is an alarm signal (the optocoupler closes when an alarm occurs).
4. PEND is the positioning signal (after positioning, the optocoupler closes).
5. IN is the input signal, IN is the zero-return switch or left limit switch, EN can be remapped to the right limit switch. For details on remapping, please refer to the limit switch setting instructions in Part 4 (when EN is mapped to the right limit switch, the pulse control mode is disabled).

② SERV057ES Interface Description



1. Input power supply 20V~48V
2. 5V is the output power (5V, 100mA).
3. ALM is an alarm signal (the optocoupler closes when an alarm occurs).
4. PEND is the positioning signal (after positioning, the optocoupler closes).
5. IN COM is the common port of IN_1 and IN_2 (can be connected in either common cathode or common anode configuration).
6. IN_1 and IN_2 are input signals. IN_1 is the zero-return switch or left limit switch, and IN_2 is the right limit switch.



1.2 Port Function Description

Port Function Table			
PIN	I/O	Function	Description
STP+ (CW+)	Input	Pulse signal (or CW pulse) positive	(1) Optocoupler input, falling edge trigger. The pulse goes one step when the pulse goes from high to low. (2) Positive voltage: 3.3 - 28V. (3) Negative voltage: High level 3.3 - 28V, Low level 0 - 0.5V. (4) Maximum input frequency 400kHz, pulse duration greater than 2.5us. (5) Single pulse (Pulse + Dir) or double pulse (CW + CCW) is set by the command.
STP- (CW-)		Pulse signal (or CW pulse) negative	
DIR+ (CCW+)		Direction signal (or CCW pulse) positive	
DIR- (CCW-)		Direction signal (or CCW pulse) negative	
EN+		Enable signal positive	(1) Optocoupler input, low level effective. When ineffective, the motor will be released while clearing the alarm signal. (2) Positive voltage: 3.3 - 28V. (3) Negative voltage: High level 3.3 - 28V, Low level 0 - 0.5V.
EN-		Enable signal negative	
ALM+	Output	Alarm signal positive	(1) When an alarm occurs, the optocoupler output is on. (2) Maximum voltage +35V, maximum current 50mA.
ALM-		Alarm signal negative	
PEND+		In Position signal positive	(1) When the drive has finished a given pulse, the optocoupler output is on. (2) Maximum voltage +35V, maximum current 50mA.
PEND-		In Position signal negative	
A-	Output	Motor phase A-	Connected to motor phase line Note: If the phase sequence is not connected accordingly, the motor will alarm.
A+		Motor phase A+	
B-		Motor phase B-	
B+		Motor phase B+	
VDC	Input	Drive power supply positive	Operating Voltage +20 - 48V
GND		Drive power supply negative	



1.3 Indicator Light Status Description Table

LED Status	
Green stays on	Motor Running
Green flash	Motor Stop
1 Green + 1 Red	Over Current
1 Green + 2 Red	Open-Phase
1 Green + 3 Red	Supply Voltage High
1 Green + 4 Red	Supply Voltage Low
1 Green + 5 Red	Position Error
1 Green + 6 Red	Encoder Error

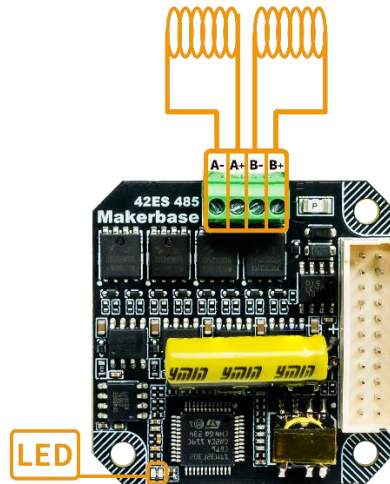
Tip: Please make sure the motor phase line is connected correctly, otherwise there will be a tracking error alarm after receiving the pulse (5 red and 1 green).

1.4 Motor wiring method

Note: The internal resistance of the motor should be less than 10 ohms.

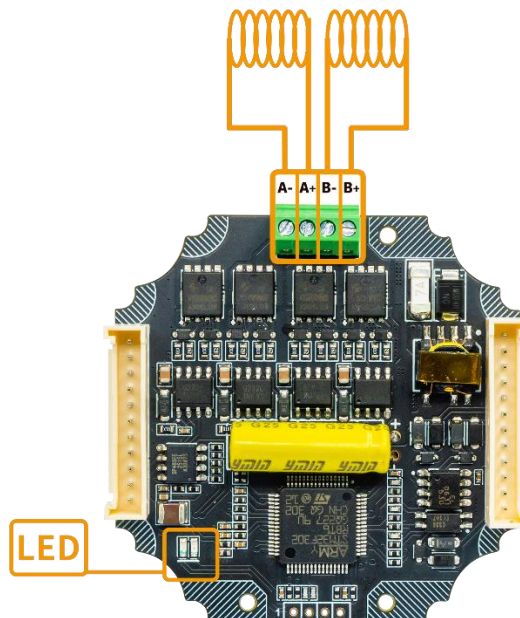
A+ and A- to one phase wire of the motor, and connect B+ and B- to the other phase wire of the motor.

① SERV042ES Wiring Method



Note: Please ensure the motor phase wires are connected correctly; otherwise, a tracking error exceeding tolerance alarm (5 red, 1 green) will occur upon receiving the pulse.

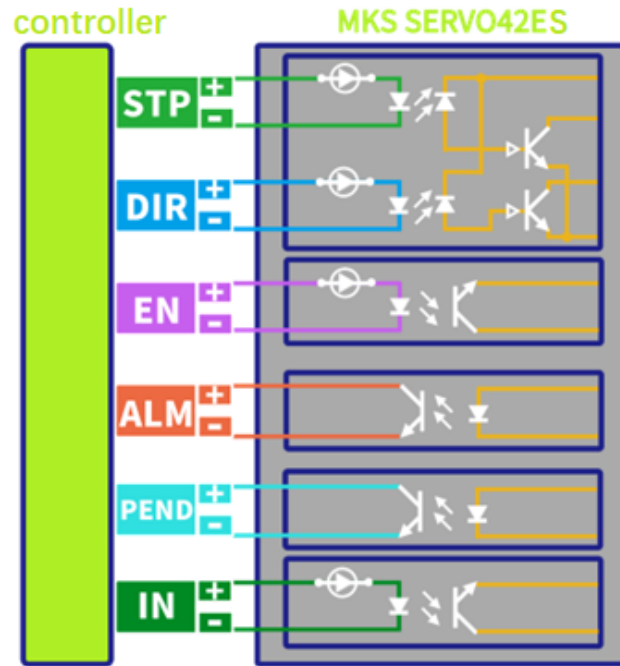
② SERV057ES Wiring Method



Note: Please ensure the motor phase wires are connected correctly; otherwise, a tracking error exceeding tolerance alarm (5 red, 1 green) will occur upon receiving the pulse.

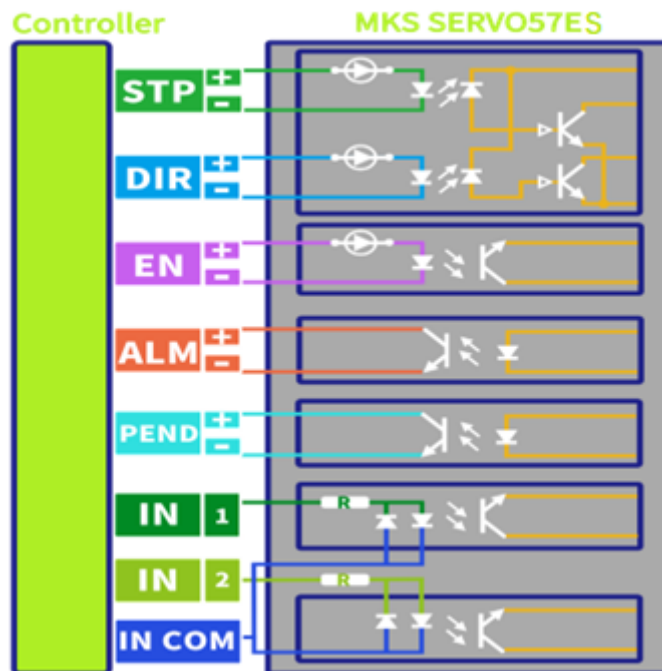
1.5 Pulse control wiring method

① SERVO42ES Wiring



Note: All input ports have a built-in 10mA current limiter, allowing direct input of 3.3V-24V signals without the need for external current-limiting resistors.

② SERVO57ES Wiring



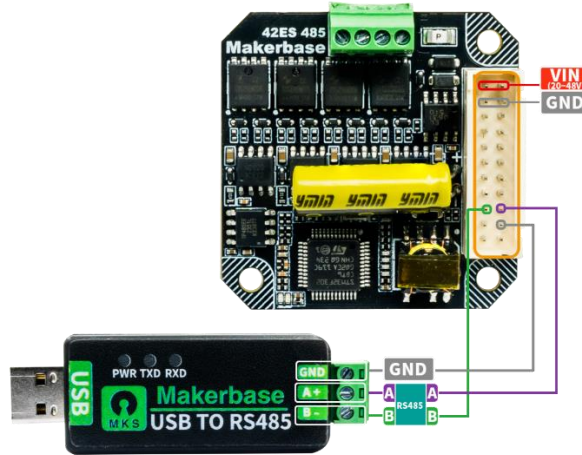
Note: All input ports have a built-in 10mA current limiter, allowing direct input of 3.3V-24V signals without the need for external current-limiting resistors.

1.6 RS485 Wiring Method

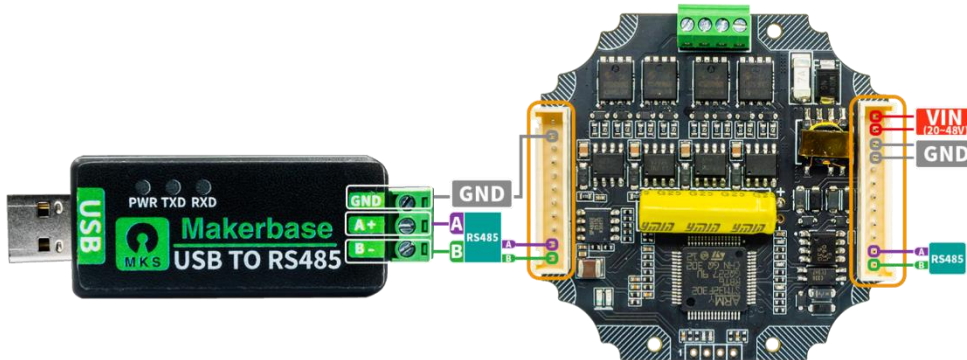
1. RS485 standalone communication wiring

Note: To reduce bus interference, the host computer and the motor must share a common ground. RS485 A and B signals should be transmitted using shielded twisted-pair cable.

① MKS SERVO42ES RS485 standalone wiring

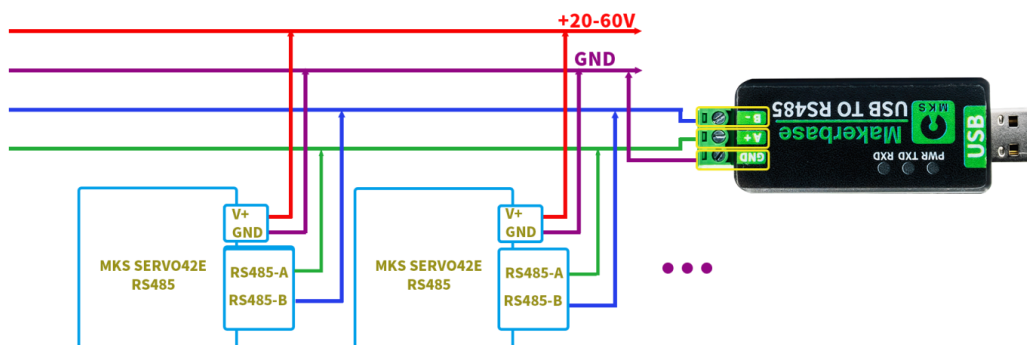


② MKS SERVO57ES RS485 standalone wiring



Note: The SERVO57ES board has two RS485 interfaces connected in parallel; connecting to either one is sufficient.

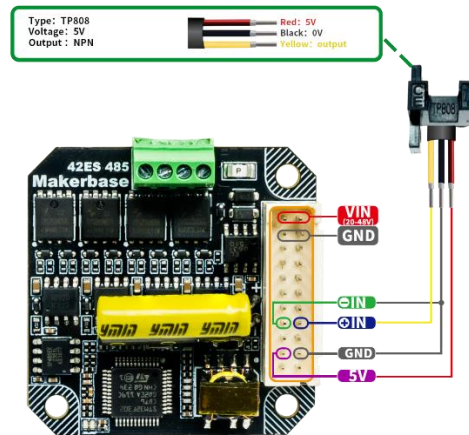
2. RS485 multi-machine communication wiring



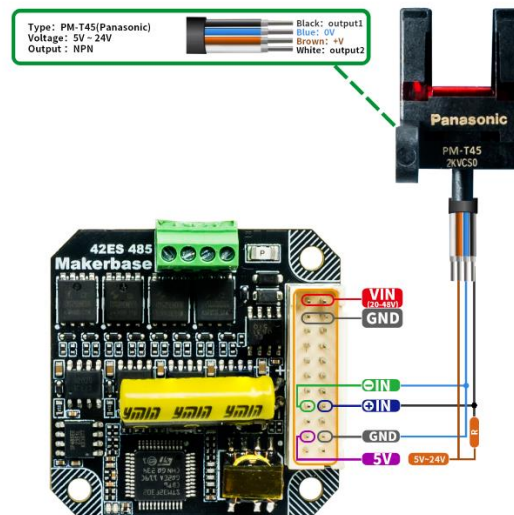
1.7 External switch wiring method

① MKS SERVO42ES Switch Wiring Method

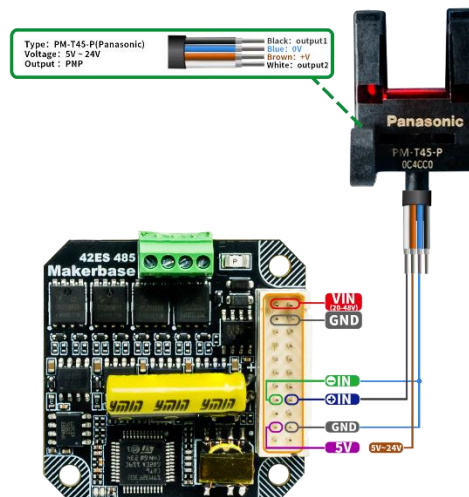
a) TP808 Wiring Method



b) PM-T45 Wiring Method

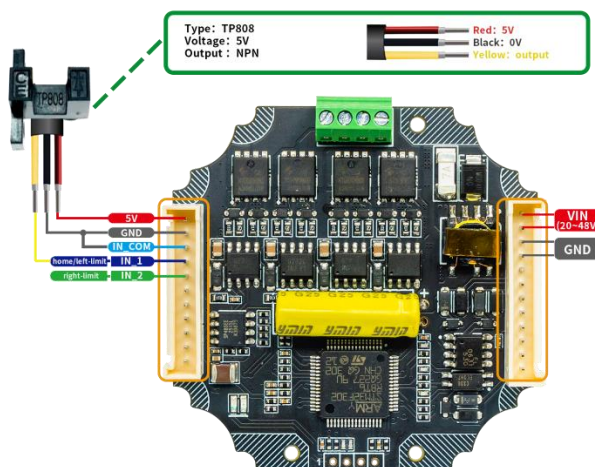


c) PM-T45-P Wiring Method

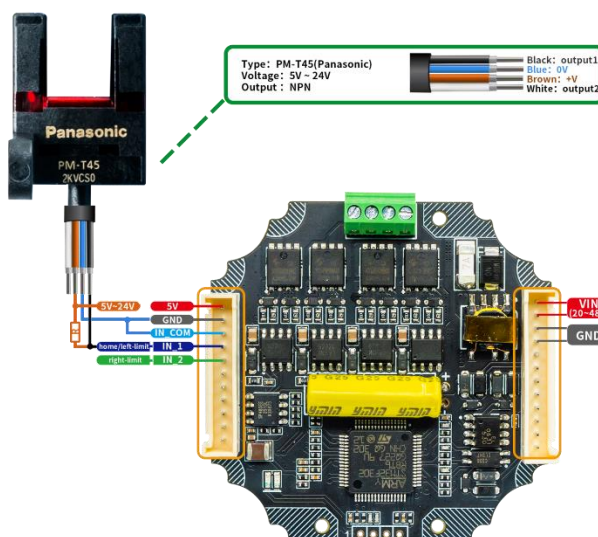


② MKS SERVO57ES Switch wiring method

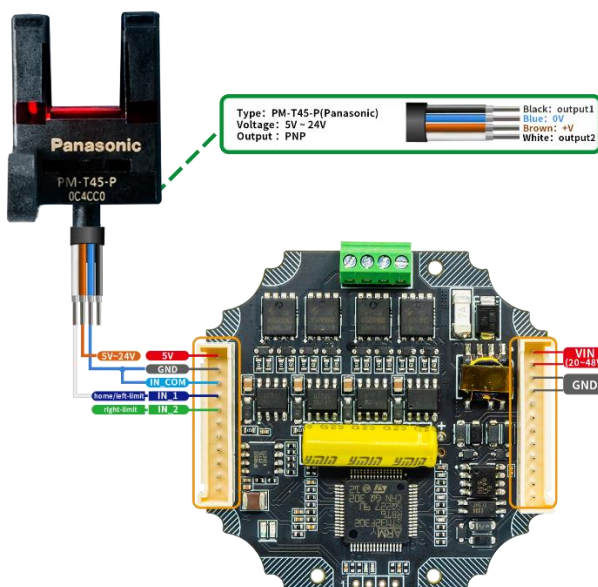
a) TP808 Wiring Method



b) PM-T45 Wiring Method



c) PM-T45-P Wiring Method





Part 2. Communication Protocol Description

2.1 RS485 bus communication

The closed-loop motor incorporates the standard Modbus RTU communication protocol, supporting the reading and writing of single and multiple parameters by the Modbus RTU master station. Once a controller with Modbus protocol is successfully connected to the closed-loop motor, the controller can directly perform parameter settings and read operations on the motor. In bus control mode, the controller can modify operating command parameters such as position and speed to control motor operation.

Please refer to the relevant documents for the Modbus-RTU/RS485 protocol and standards.

2.2 Communication parameters

The default baud rate is 38400, with 8 data bits, none parity bits, and 1 stop bit.

2.3 Slave address

The default address is 01, and the value can be set from 00 to 255.

Note: 00 is the broadcast address. When using the broadcast address, the slave device will not respond.

2.4 code

The Modbus RTU protocol has a variety of bus commands. Closed-loop motors support the three most commonly used code commands (03H/06H/10H). These three codes can meet all the controller's control needs for the closed-loop motor.

The Modbus register has a 16-bit data length. When using Modbus commands, pay attention to the data type of the access parameters .

The parameter data type is UINT16. INT16 should be read using code 03H and written using code 06H.

For parameters of data type UINT32 or INT32, or for reading or writing multiple parameters consecutively, use codes 03H for reading and 10H for writing.



2.4.1 03H (Read a single register)

Use function code 03H to read only one parameter. The starting address of the register is the register address of that parameter, and the returned data is the data corresponding to that parameter.

For example, if the controller wants to read the real-time speed of the motor (register address 06H), it can send the following query data frame:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	06H	00H	01H	64H	0BH

If the real-time motor speed is 0x012CRPM, the response frame will be as follows:

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	02H	01H	2CH	B8H	09H



2.4.2 03H (Read multiple registers)

Some parameters are 32-bit data types, while others are 16-bit data types. To read multiple parameters consecutively, you need to use the 03H code to read multiple consecutive 16-bit registers.

When reading multiple consecutive parameters using 03H, the register's starting address is the address of the first parameter's register. The returned consecutive data order is: first parameter data → second parameter data → Nth parameter data.

The system automatically offsets sequentially based on the first returned parameter data, and the amount of returned data depends on the Numbers read.

For example, if the controller wants to read the high 32 bits of the motor encoder position (register address 04H) and the real-time motor speed (register address 06H), it can send the following query data frame:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	04H	00H	03H	44H	0AH

Assuming the encoder position high 32 bits are 0x12345678 and the motor real-time speed is 0x0258 RPM, the response frame is as follows:

Uplink message (host computer ← SERVOxxES)										
Address	code	Number of bytes	Address 4 data		Address 5 data		Address 6 data		CRC16	
			H	L	H	L	H	L	H	L
01H	03H	06H	56H	78H	12H	34H	02H	58H	C9H	65H



2.4.3 06H (Write a single register)

function Code 06H can only write one parameter with a length of 16 bits. The register starting address is the register address of this parameter. After the closed-loop motor successfully receives the query data frame, it changes the value of this parameter to the data to be written.

For example, if the controller wants to write the operating current (register address 11H) as 1600mA, it can send the following query data frame.

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	11H	06H	40H	64H	0BH

If the write is successful, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	11H	06H	40H	64H	0BH

If the write operation fails, the response frame will look like this:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	11H	FFH	FFH	64H	0BH

The reasons for write failure include:

This address does not support writing.

The data written exceeds the specified range.

Note: After all parameters are written, a save command (address 54H) must be executed once for the written parameters to be saved after power failure.



2.4.4 10H (Write to multiple registers)

Some parameters are 32-bit data types, while others are 16-bit data types. To write multiple parameters consecutively, you need to use the 10H code to write multiple consecutive 16-bit registers.

When writing multiple consecutive parameters using register 10H, the starting address of the register is the address of the first parameter. The order of writing consecutive data is: first parameter data → second parameter data → Nth parameter data.

For example: the controller wants to write...

The zero-origin offset (telephone address 2EH) is 0x23456789.

The zero-return current (register address 30H) value is 0x0258.

The following query data frames can be sent:

Downlink message (host computer → SERV0xxES)														
addr	code	Start addr		Reg Num		bytes	2EH		2FH		30H		校验码	
		H	L	H	L		H	L	H	L	H	L	H	L
01H	10H	00H	2EH	00H	03H	06H	67H	89H	23H	45H	02H	58H	49H	1CH

If the write is successful, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	2EH	00H	03H	E0H	01H

If the write operation fails, the response frame will look like this:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	2EH	00H	00H	A0H	00H

The reasons for write failure include:

This address does not support writing.

The written data exceeded the specified range.

Number of bytes written $\neq 2 * \text{Numbers}$

Note: After all parameters are written, a save command (address 54H) must be executed once for the written parameters to be saved after power failure.



Part 3. Register List

address	Description	Accessibility	type	scope	default	unit	active
0000H	Hardware version	RO	UINT16	—		—	—
0001H	Firmware version	RO	UINT16	—		—	—
0002H	Encoder value (lower 32 bits)	RO	INT32	—	—	Encoder unit	—
0003H							
0004H	Encoder value (high 32 bits)	RO	INT32	—	—	Encoder unit	—
0005H							
0006H	Real-time motor speed	RO	INT16	—	—	RPM	—
0007H	Motor alarm status	RO	UINT16	—	—	—	—
0008H	Number of received pulses	RO	INT32	—	—	—	—
0009H							
000AH	Position angle error	RO	INT32	—	—	—	—
000BH							
000CH	Motor enable state	RO	UINT16	—	—	—	—
000DH	Motor return to zero state	RO	UINT16	—	—	—	—
000EH	reserve						
000FH	reserve						
0010H	Work mode	R/W	UINT16	0-5	3	—	Reboot
0011H	Operating current	R/W	UINT16			Ma	immediately
0012H	Job breakdown	R/W	UINT16		16	—	immediately
0013H	EN active level	R/W	UINT16	0-2	0	—	immediately
0014H	Rotation direction	R/W	UINT16	0-1	0	—	immediately
0015H	Pulse delay	R/W	UINT16	0-3	2	ms	immediately
0016H	baud rate	R/W	UINT16	1-7	4	—	Reboot
0017H	Slave address	R/W	UINT16	0-255	1	—	Reboot
0018H	Protocol Mode	R/W	UINT16	0-1	0	—	immediately
0019H	Does bus startup lock the axis?	R/W	UINT16	0-1	1	—	immediately
001AH	reserve		UINT16				
001BH	reserve		UINT16				
001CH	Stall protection	R/W	UINT16	0-1	1	—	Reboot
001DH	Excessive stall deviation	R/W	UINT16	0-0x7FFF	0x64	Encoder unit	Reboot
001EH	Motor stall state	RO	UINT16	0-1		—	—
001FH	Release motor stall	WO	UINT16	1		—	immediately
0020H	reserve		UINT16				
0021H	reserve		UINT16				
0022H	IO port read	RO	UINT16			—	—
0023H	Write to IO port 1	WO	UINT16			—	immediately
0024H	Write to IO port 2	WO	UINT16			—	immediately



address	Description	Accessibility	type	scope	default	unit	active
0025H	Pulse frequency division level reversal	R/W	UINT16	0-1	0	–	Reboot
0026H	Pulse frequency division period	R/W	UINT32		0	–	Reboot
0027H							
0028H	reserve		UINT16				
0029H	reserve		UINT16				
002AH	Zero mode	R/W	UINT16	0-3	0	–	immediately
002BH	Return to zero direction	R/W	UINT16	0-1	0	–	immediately
002CH	Return to zero "high speed" speed	R/W	UINT16	1-3000	100	RPM	immediately
002DH	Return to zero "low speed" speed	R/W	UINT16	1-100	10	RPM	immediately
002EH	Zero origin offset	R/W	UINT32		0x2000	Encoder unit	immediately
002FH							
0030H	Zero-return current	R/W	UINT16			mA	immediately
0031H	Zero-return trigger method	R/W	UINT16	0-2	0	–	immediately
0032H	Zeroing timeout	R/W	UINT32		0x1D4C0	ms	immediately
0033H							
0034H	Set "Zero Point"	WO	UINT16	1		–	immediately
0035H	Execute back to zero	WO	UINT16	0-1		–	
0036H	reserve		UINT16				
0037H	Limit switch active/inactive	R/W	UINT16	0-1	0	–	immediately
0038H	IO input valid level	R/W	UINT16	0-1	0	–	immediately
0039H	Right limit mapping	R/W	UINT16	0-1	0	–	immediately
003AH	reserve		UINT16				
003BH	reserve		UINT16				
003CH	Motor operating status	RO	UINT16			–	
003DH	Motor enable	WO	UINT16	0-1		–	immediately
003EH	Emergency stop of motor	WO	UINT16	1		–	immediately
003FH	reserve		UINT16				



address	Description	Accessibility	type	scope	default	unit	active
0040H	Automatic operation upon power-on	R/W	UINT16	0-1	0	–	Reboot
0041H	Target speed	R/W	UINT16	1-3000	0	RPM	immediately
0042H	Running target acceleration	R/W	UINT16	0-255	0	–	immediately
0043H	Direction of movement	R/W	UINT16	0-1	0	–	immediately
0044H	runtime	R/W	UINT32		0	10ms	immediately
0045H							
0046H	Speed Run Command	WO	UINT16	0-1		–	immediately
0047H	reserve		UINT16				
0048H	Number of running pulses	R/W	INT32		0	–	immediately
0049H							
004AH	Pulse running command	WO	UINT16	0-1		–	immediately
004BH	reserve	WO	UINT16			–	immediately
004CH	Running coordinates	R/W	INT32		0	Encoder unit	immediately
004DH							
004EH	Coordinate running instructions	WO	UINT16	0-1		–	immediately
004FH	reserve	WO	UINT16			–	immediately
0050H	reserve		UINT16				
0051H	reserve		UINT16				
0052H	reserve		UINT16				
0053H	reserve		UINT16				
0054H	Save parameter command	WO	UINT16	1		–	immediately
0055H	Restart the motor	WO	UINT16	1		–	immediately
0056H	Restore factory settings	WO	UINT16	1		–	immediately
0057H	Calibrate the motor	WO	UINT16	1			immediately
0058H	User-defined ID	R/W	UINT32	–	0	–	immediately
0059H							
005AH	IAP Firmware Upgrade	WO	UINT16	0-1		–	immediately



Part 4. General Instructions

4.1 Calibrate motor parameters

Note: The complete machines sold by the manufacturer are already calibrated at the factory and do not require recalibration.

Calibration can compensate for and correct nonlinearities caused by magnet installation deviations.

If the motor phase wires are wired according to the factory default connection, no calibration is required. Although it can be used, the motor performance may be poor.

To ensure the motor performs optimally, or if issues such as excessive motor vibration or inconsistent current (sound) in both forward and reverse directions occur, the motor can be recalibrated.

The motor must be unloaded before calibration and must not be under any load.

During calibration, the motor rotates at a speed of 600 RPM for more than 20 revolutions.

If calibration is successful, the green light will flash rapidly. Power off and restart the motor before it can be used normally.

If calibration fails, the red light will flash rapidly. You need to power off and restart the motor, and then calibrate again.

Register address: 57H

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	57H	00H	01H	F9H	DAH

No data was returned.

After receiving the command, the motor will automatically restart and enter calibration mode. You need to wait about 10 seconds to observe the status of the LED lights and judge the calibration result.



4.2 Read parameter command

4.2.1 Read hardware version

Address: 00H

Data type: uint16_t

Note: Read hardware version

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	00H	00H	01H	84H	0AH

Uplink message (host computer ← SERV0xxES)					
Address	code	Number of bytes	Register data		CRC16
			H	L	H L
01H	03H	02H	Hardware version		

4.2.2 Read firmware version

Address: 01H

Data type: uint16_t

Note: Read firmware version

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	01H	00H	01H	D5H	CAH

Uplink message (host computer ← SERV0xxES)					
Address	code	Number of bytes	Register data		CRC16
			H	L	H L
01H	03H	02H	Firmware version		



4.2.3 Read the lower 32 bits of the encoder value

Address: 02H – 03H

Data type: int32_t

Explanation: Combined with the high 32 bits, this forms a 64-bit encoder value, recording the motor's rotation range after power-on (enabled or disabled).

Encoder single-turn value range: 0~0x4000

When moving relative to or absolutely according to coordinate values, use the encoder value as the coordinate.

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	02H	00H	02H	65H	CBH

Assuming the encoder's lower 32 bits are 0x12345678, the returned data format is as follows:

Uplink message (host computer ← SERV0xxES)								
Address	code	Number of bytes	02H data		03H data		CRC16	
			H	L	H	L	H	L
01H	03H	04H	56H	78H	12H	34H	66H	D5H

4.2.4 Read the high 32 bits of the encoder value

Address: 04H – 05H

Data type: int32_t

Explanation: Combined with the lower 32 bits, this forms a 64-bit encoder value, recording the motor's rotation range after power-on (enabled or disabled).

Encoder single-turn value range: 0~0x4000

When moving relative to or absolutely according to coordinate values, use the encoder value as the coordinate.

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	04H	00H	02H	85H	CAH

Assuming the encoder's high 32 bits are 0x12345678, the returned data format is as follows:

Uplink message (host computer ← SERV0xxES)								
Address	code	Number of bytes	04H data		05H data		CRC16	
			H	L	H	L	H	L
01H	03H	04H	56H	78H	12H	34H	66H	D5H



4.2.5 Read the real-time speed of the motor

Address: 06H

Data type: int16_t

Note: Real-time motor speed, unit RPM

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	06H	00H	01H	64H	0BH

Assuming the motor's real-time speed is 0x012CRPM, the response frame would be as follows:

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	02H	01H	2CH	B8H	09H

4.2.6 Read motor alarm status

Address: 07H

Data type: uint16_t

Note: Motor alarm status

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	07H	00H	01H	35H	CBH

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	02H		status		

Motor status	status	Remark
motor running	0	
Motor stopped	1	
Over current error	2	
Phase loss error	3	
Overvoltage error	4	
Undervoltage error	5	
Position error	6	
Encoder error	7	



4.2.7 Read the number of received pulses

Address: 08H – 09H

Data type: int32_t

illustrate: Number of received pulses

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	08H	00H	02H	45H	C9H

Assuming the number of received pulses is 0x12345678, the returned data format is as follows:

Uplink message (host computer ← SERV0xxES)								
Address	code	Number of bytes	08H data		09H data		CRC16	
			H	L	H	L	H	L
01H	03H	04H	56H	78H	12H	34H	66H	D5H

4.2.8 Position angle error

Address: 0AH – 0BH

Data type: int32_t

Explanation: Angle error is the difference between the position angle controlled by the command and the real-time angle position of the motor.

Unit: 0~51200 represents 0~360°. For example, when the error is 1°, the value is 51200/360° = 142.22, and so on.

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	0AH	00H	02H	E4H	09H

Assuming the error is 0x12345678, the returned data format is as follows:

Uplink message (host computer ← SERV0xxES)								
Address	code	Number of bytes	0AH data		0BH data		CRC16	
			H	L	H	L	H	L
01H	03H	04H	56H	78H	12H	34H	66H	D5H



4.2.9 Motor enable state

Address: 0CH

Data type: uint16_t

Note: Obtain the motor's enable status: 00 Disabled / 01 Enabled

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	0CH	00H	01H	44H	09H

Assuming the motor is enabled, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	02H	00H	01H	79H	84H

4.2.10 Motor return to zero state

Address: 0DH

Data type: uint16_t

Note: Obtain the motor's zero-return status: 00 Not zeroed / 01 Zeroed

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	0DH	00H	01H	15H	C9H

Assuming the motor has not returned to zero, the response frame will be as follows:

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	02H	00H	00H	B8H	44H



4.3 Configuration parameter commands

Note: After all parameters are configured, a save command (address 54H) must be executed once for the configured parameters to be saved after power failure.

4.3.1 Work mode

Address: 10H

Note: Configure motor operating mode

Data range: See the working mode configuration table.

Assuming the operating mode is configured as "RS485 bus closed-loop FOC mode", the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	10H	00H	05H	48H	0CH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	10H	00H	05H	48H	0CH

Note: If the written data exceeds the data range, the write operation will be successful, but the actual configuration will fail.

The corresponding working modes are shown in the table below.

List of working modes		
	mode	Work mode
With encoder	00H	Pulse + Pulse Open-Loop FOC Mode
	01H	Pulse + Direction Open-Loop FOC Mode
	02H	Pulse + Pulse Closed-Loop FOC Mode
	03H	Pulse + Direction Closed-Loop FOC Mode (Default)
	04H	RS485 bus open-loop FOC mode
	05H	RS485 bus closed-loop FOC mode
No encoder	10H	Pulse + Pulse Open-Loop FOC Mode
	11H	Pulse + Direction Open-Loop FOC Mode
	14H	RS485 bus open-loop FOC mode

Note 1: An encoder is present, meaning the motor shaft has magnets. The drive board is mounted on the back and can read the encoder values.

No encoder, meaning the motor shaft has no magnets, the drive board can be installed arbitrarily, and the encoder value cannot be read.

Note 2: Pulse control mode, maximum input frequency 300kHz

Bus control mode, maximum speed 3000 RPM



4.3.2 Operating current

Address: 11H

Data type: uint16_t

Note: Configure motor operating current

Unit: milliampere (mA)

The maximum current of the 57ES is 5200mA (default 3200mA).

The maximum current of the 42ES is 3000mA (default 1600mA).

Assuming the operating current is configured to be 800mA, the query frame would be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	start address		Data		CRC16	
		high	low	high	low	high	low
01H	06H	00H	11H	03H	20H	D8H	E7H

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	start address		Data		CRC16	
		high	low	high	low	high	low
01H	06H	00H	11H	03H	20H	D8H	E7H

Note: If the written data exceeds the maximum value, the motor operating current will be limited to the maximum value.

4.3.3 MicSteps

Address: 12H

Note: Configure motor microstepping

Data type: uint16_t

Data range: The subdivided value range (decimal) is as follows: (default value 16)

0,2,4,8,16,32,64,128, 5,10,20,25,40,50,100,200

Note: 0 corresponds to 256 subdivisions. Data outside this range will not be configured successfully.

Assuming the configuration task is subdivided into 32, the query frame would be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	start address		Data		CRC16	
		high	low	high	low	high	low
01H	06H	00H	12H	00H	20H	28H	17H

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	start address		Data		CRC16	
		high	low	high	low	high	low
01H	06H	00H	12H	00H	20H	28H	17H

Note: Data outside the configuration range may be successfully written, but the actual configuration will fail.



4.3.4 EN active level

Address: 13H

Note: Configure the motor's EN active level.

Data type: uint16_t

Data range:

00 corresponds to low-level enable (L) (default value)

01 corresponds to high-level enable (H)

02 corresponds to always being enabled (Hold)

Assuming the configuration is low-level enable, the query frame would be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	13H	00H	00H	78H	0FH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	13H	00H	00H	78H	0FH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.

4.3.5 Rotation direction

Address: 14H

Note: Configure the motor rotation direction

Data type: uint16_t

Data range:

00 corresponds to clockwise rotation (default value)

01 corresponds to counterclockwise rotation

Assuming the configuration is clockwise, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	14H	00H	00H	C9H	CEH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	14H	00H	00H	C9H	CEH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



4.3.6 Pulse delay

Address: 15H

Note: Configure delay parameters

Data type: uint16_t

Data range:

- | | | | |
|----|----------------------------|----|------------|
| 00 | Delay 0ms | 01 | Delay 4ms |
| 02 | Delay 20ms (default value) | 03 | Delay 40ms |

Assuming a 4ms delay, the query frame would look like this:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	15H	00H	01H	59H	CEH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	15H	00H	01H	59H	CEH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.

4.3.7 baud rate

Address: 16H

Note: Configure the baud rate. After saving the configuration parameters, a reset and restart are required for the changes to take effect.

Data type: uint16_t

Data range:

- | | | | |
|----|------------------|----|---------------------------|
| 01 | Baud rate 9600 | 02 | Baud rate 19200 |
| 03 | Baud rate 25000 | 04 | Baud rate 38400 (Default) |
| 05 | baud rate 57600 | 06 | baud rate 115200 |
| 07 | baud rate 256000 | | |

Assuming a baud rate of 115200, the query frame would be as follows.

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	16H	00H	06H	E8H	0CH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	16H	00H	06H	E8H	0CH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



4.3.8 Slave address

Address: 17H

Note: Configure the slave address; the configuration will take effect immediately.

Data type: uint16_t

Data range: 00-0xFF, default address 01, broadcast address 00

Assuming the slave address is 02, the query frame would be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	17H	00H	02H	B8H	0FH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	17H	00H	02H	B8H	0FH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.

4.3.9 Protocol Mode

Address: 18H

Note: Select the Modbus RTU protocol or MKS custom protocol, configure and save the parameters, then reset and restart for the changes to take effect.

Data type: uint16_t

Data range: 00 MKS custom protocol (default value)

01 Modbus RTU Protocol

Assuming the configuration is a custom MKS protocol, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	18H	00H	00H	09H	CDH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	18H	00H	00H	09H	CDH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.

Note: In MKS custom protocol mode, to switch to Modbus RTU protocol mode, use a serial port assistant to send the following command.

FA 01 8E 01 8A (Configure Modbus RTU protocol)

FA 01 60 01 5C (Save parameters)

FA 01 41 3C (Reset and restart motor)

After a reset and restart, it will enter Modbus RTU protocol mode.



4.3.10 Locking shaft

Address: 19H

Note: When starting in bus mode, is the motor shaft locked?

Data type: uint16_t

Data range:

- 00 Start without locking shaft
- 01. Activate shaft locking (default value)

Assuming the configuration is set to enable axis locking, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	19H	00H	01H	99H	CDH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	19H	00H	01H	99H	CDH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



4.4 Stall protection instructions

If the motor fails to reach the designated position within the response time due to overload or any other reason, the stall protection will be triggered, the motor will unlock and the shaft will be released to avoid damage.

4.4.1 Configure stall protection

Address: 1CH

Note: Configure stall protection function.

Data type: uint16_t

Data range: 00 Stall protection function disabled.

01 Enable stall protection function (default value)

Assuming the configuration is: stall protection is enabled, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	1CH	00H	01H	89H	CCH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	1CH	00H	01H	89H	CCH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.

4.4.2 Configure stall tolerance

Address: 1DH

Note: An out-of-tolerance value is configured. When the error exceeds the out-of-tolerance value, the stall protection is triggered, and the motor shaft is released.

Data type: uint16_t

Data range: 0 – 0x7FFF, default value 0x64

Assuming a configuration error of 0xC8, the query frame would be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	1DH	00H	C8H	18H	5AH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	1DH	00H	C8H	18H	5AH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



4.4.3 Read motor stall status

When a motor stalls, a stall flag will be set. This command allows you to determine if the motor has stalled. If the stall protection option is enabled, the motor will automatically unlock after a stall occurs.

Address: 1EH

Data type: uint16_t

Note: Obtain the stall status of the motor: 00 Not stalled / 01 Stalled

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	1EH	00H	01H	E4H	0CH

Assuming the motor is stalled, the response frame would be as follows:

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	02H	00H	01H	79H	84H

4.4.4 Release the motor from stall state

Address: 1FH

Note: Release stall protection

Data type: uint16_t

Assuming the configuration is: disable stall protection, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	1FH	00H	01H	79H	CCH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	1FH	00H	01H	79H	CCH

When the motor stalls, sending this command can release the current stall state.

If a stall occurs again after the stall is cleared , the stall protection will still be triggered.

Note: controlling the motor shaft to loosen can also release the stall state .



4.5 Save /Restore Parameters and Reset Instructions

4.5.1 Save parameter command

Address: 54H

Note: All modified parameters are saved to internal storage.

Data type: uint16_t

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	54H	00H	01H	09H	DAH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	54H	00H	01H	09H	DAH

4.5.2 Reset and restart the motor

Address: 55H

Note: Reset and restart the motor.

Data type: uint16_t

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	55H	00H	01H	58H	1AH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	55H	00H	01H	58H	1AH



4.5.3 Restore factory settings

Address: 56H

Note: Restore factory default parameters and restart the motor.

Data type: uint16_t

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	56H	00H	01H	A8H	1AH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	56H	00H	01H	A8H	1AH



4.6 IAP Firmware Online Upgrade Instructions

Address: 5AH

Note: The motor has entered boot mode and is awaiting firmware upgrade.

Data type: uint16_t

Data range: 01 Enter boot mode, retain user parameters

If: Entering boot mode and restoring factory settings, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	5AH	00H	01H	68H	19H

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	5AH	00H	01H	68H	19H

After successful configuration, the motor will automatically restart and enter boot mode, waiting to receive the upgrade file.

For detailed instructions, please refer to the following documents or videos:

"MKS SERV042&57E IAP Upgrade Instructions.pdf".

"MKS SERV042&57E IAP Upgrade Operation Video.mp4".



Part 5. IO Port Operation Instructions

5.1 Read I/O port status

Address: 22H

Data type: uint16_t

Note: Read IO port status

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	22H	00H	01H	24H	00H

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	22H	00H	status		

IO Port Status Definition Table

status							
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Undefined				ALM	PEND	IN_2	IN_1

PEND 1: In place; 0: Not in place

ALM 1: No alarm; 0: Alarmed

Note: 42ES does not have the IN_2 port; bit1 corresponds to the state of the En port.



5.2 Write ALM port data

Address: 23H

Note: Write to ALM port

Data type: uint16_t

mask data range 0: Do not write to the ALM port (ALM default alarm signal)

1: Write the data value to the ALM port.

2: The ALM port value remains unchanged.

The data range is 0/1 (writing 1 to the ALM port disconnects the corresponding optocoupler).

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	23H	mask	data		

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	23H	mask	data		

Note: Although the data was written successfully, the actual configuration failed because it was out of range.

5.3 Write PEND port data

Address: 24H

Note: Write PEND port

Data type: uint16_t

mask data range 0: No data is written to the PEND port (PEND is a default bit signal).

1: Write the data value to the PEND port.

2: The PEND port value remains unchanged.

The data range is 0/1 (writing 1 to the PEND port will close the corresponding optocoupler).

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	24H	mask	data		

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	24H				

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



5.4 Pulse frequency division output

Address: 25H-27H

Note: The PEND port outputs a pulse frequency division signal.

Data types: uint16_t + uint32_t

divLevel defines the output start level (default value 0).

0: Start-low level

1: Start-up high level

divPeriod defines the frequency division period (default value 0, meaning no frequency division signal is output).

When divPeriod < 100, no frequency divider output

When divPeriod >= 100, the PEND port toggles once every divPeriod pulse cycle.

For example, setting 16 subdivisions, divPeriod = 3200, then the PEND port flips once for every revolution of the motor.

The query frame is as follows:

Downlink message (host computer → SERV0xxES)												
Address	code	Starting address		Numbers		Number of bytes written	25H data		26H data		27H data	
		high level	low position	high level	low position		high level	low position	high level	low position	high level	low position
01H	10H	00H	25H	00H	03H	06H	divLevel		divPeriod			

If the write is successful, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	25H	00H	03H	91H	C3H



Part 6. Motor return to zero instructions

There are two types of motor homing methods: "origin homing" and "coordinate homing".

6.1 Explanation of the method of returning to zero at the origin

There are two ways to "return to zero": "return to zero by origin switch" and "return to zero by mechanical limit switch".

1. Return the origin switch to zero.

Connect the origin switch to the corresponding port:

Origin switch access port	
SERVO57ES	IN_COM / IN1
SERVO42ES	IN+ / IN-

For detailed wiring instructions, please refer to 1.7 External switch wiring method.

The process of returning to zero is as follows:

- The motor first searches for the switch in the set "direction" and "high speed " ;
- Upon encountering the rising edge of the switch signal, it begins to operate at "low speed " and disengages from the switch;
- After the switch is disconnected, the motor reverses direction and continues to search for the rising edge of the switch signal at "low speed " ;
- Upon encountering the rising edge of the switch signal, it then runs at "high speed " to the preset "origin offset" position and stops.
- Mark the current position as "coordinate zero point", and the origin has successfully returned to zero.

2. Mechanical limit switch returns to zero.

The zero-return torque when the mechanical limit is returned to zero needs to be set in advance via command 94H. The set torque should be sufficient to drive the load, but should not be too large to avoid damaging the equipment.

The process of returning to zero is as follows:

- searches for the mechanical limit position using the set "direction", "torque " , and "low speed " ;
- When encountering a mechanical limit switch, the rotor will stop and then run to the preset "origin offset" position to stop.
- Mark the current position as "coordinate zero point", and the origin has successfully returned to zero.



3. Single lap to zero

The "zero point" of the coordinates within a single circle needs to be set in advance via command 92H.

The direction of the single-turn zeroing needs to be set in advance via command 93H: "forward", "reverse", or "nearest".

The process of returning to zero is as follows:

- a. The motor returns to the "coordinate zero point" position within a preset single revolution at high speed in the set zero-return direction;
- b. Upon arrival, the current position is reset to zero, and the single-lap coordinates have successfully returned to zero.

6.2 Explanation of coordinate zeroing method

To return to zero directly, you need to first execute the "Return to Zero" function to determine the "zero point".

The process of returning to zero is as follows:

No search process is required; it directly and quickly reaches the "coordinate zero point" position and successfully returns to zero.



6.3 Set parameters related to zero return.

6.3.1 Set the zero-return mode, direction, and speed parameters.

Address: 2AH-2DH

Instructions: Set the return-to-zero mode, direction, and speed.

Data type: uint16_t

Data Explanation:

Hm Mode: Sets the zero-return mode

00: Origin switch returns to zero (default value)

01: Mechanical limit switch returns to zero

02: Single lap zeroing

03: Disable the zero-reset function

Hm Dir: Sets the return-to-zero direction

00: Positive zero return (default value)

01: Reverse return to zero

02: Zero at the nearest point (only applicable to zeroing on a single lap)

HiSpeed: Sets the "high-speed" RPM for homing (range 1~3000 RPM, default value = 100).

LoSpeed: Sets the "low speed" RPM when returning to zero (range 1~100RPM, default value = 10).

The query frame is as follows:

Downlink message (host computer → SERV0xxES)																
Addr	code	Start address		register quantity		bytes	2AH data		2BH data		2CH data		2DH data		CRC16	
		H	L	H	L		H	L	H	L	H	L	H	L	H	L
01H	10H	00H	2AH	00H	04H	08H	Hm Mode		Hm Dir		HiSpeed		LoSpeed			

If the write is successful, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	2AH	00H	04H	E0H	02H

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



6.3.2 Set the zero-origin offset and torque parameters.

Address: 2EH-30H

Note: Set the zero-origin offset and torque parameters.

Data types: uint32_t / uint16_t

Data Explanation:

Hmoffset: Sets the offset position after returning to zero (default value = 0x2000, i.e., half a circle).

When Hmoffset = 0, the motor returns to zero and then stops.

When Hmoffset $\neq$ 0, after the motor returns to zero, it will continue running until it reaches the offset position before stopping.

Note: The Orgoffset value is invalid when lapping back to zero.

HmTorque : Sets the current when the mechanical limit switch returns to zero.

The SERV042ES has a maximum return-to-zero current of 3000mA (default 300mA).

The SERV057ES has a maximum return-to-zero current of 5200mA (default 600mA).

The query frame is as follows:

Downlink message (host computer → SERV0xxES)														
addr	code	Start address		register quantity		bytes	2EH		2FH		30H		CRC16	
		H	L	H	L		H	L	H	L	H	L		
01H	10H	00H	2EH	00H	03H	06H	Hmoffset		HmTorque					

If the write is successful, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	2EH	00H	03H	E0H	01H



6.3.3 Configure the zero-return trigger method and timeout parameters.

Address: 31H-33H

Note: Set the zero-return trigger method and timeout parameters.

Data types: uint16_t / uint32_t

Data Explanation:

Hm Trig: Sets the zero-return trigger method

00: Command triggers reset to zero (default value)

01: Automatic return to zero upon power-on (only valid in bus control mode)

02: En signal triggers zero return (only valid in pulse control mode)

In pulse control mode, when hmTrig = 2, the motor is triggered to return to zero when the En signal line generates a 200ms-wide disabled level. (Pulse recognition range: 150ms~250ms)

When En is low, it enables the motor and generates a 200ms high-level signal, triggering the motor to return to zero.

When En is high, it enables the motor and generates a 200ms low-level signal, triggering the motor to return to zero.

Hm TimOut: Sets the timeout period for returning to zero (default value = 120000).

If the motor does not reach zero within this time, a timeout failure message will be returned.

The query frame is as follows:

Downlink message (host computer → SERV0xxES)														
addr	code	Start address		register quantity		bytes	31H		32H		33H		CRC16	
		H	L	H	L		H	L	H	L	H	L	H	L
01H	10H	00H	31H	00H	03H	06H	HmTrig		HmTimOut					

If the write is successful, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	31H	00H	03H	D1H	C7H

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



6.3.4 Set "Zero Point" command

Address: 34H

Note: Set the current position directly to "zero point".

Data type: uint16_t

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	34H	00H	01H	09H	C4H

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	34H	00H	01H	09H	C4H

6.3.5 Execute the zero-return instruction

Address: 35H

Note: Motor search "zero point"

Data type: uint16_t

Data Explanation: 00: Execute the "Return to Zero" function.
 01: Execute the "Coordinate Zeroing" function

Note: You can execute "Return to Zero" first to determine the "Zero Point Coordinates" before executing "Return to Zero".

If the "return to zero" command is executed, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	35H	00H	00H	99H	C4H

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	35H	00H	00H	99H	C4H



6.4 Example of Switch return to zero configuration

Connect the origin switch to the corresponding port, referring to "

6.1 Explanation of the method of returning to zero at the origin"

6.4.1 Example of a switch command triggering a return to zero

The setup steps are as follows:

1. Configure **working mode**
01 06 00 10 00 05 48 0C
2. Configure **the active level of the switch I/O input**
01 06 00 38 00 00 08 07
3. Configure **return-to-zero mode, direction, "High Speed", "Low Speed"**
01 10 00 2A 00 04 08 00 00 00 00 00 64 00 0A EE 05
4. Configure **origin offset**
01 10 00 2E 00 02 04 20 00 00 00 7B FB
5. Configure the zero-return **trigger method** and **timeout parameters**.
01 10 00 31 00 03 06 00 00 EA 60 00 00 83 BC
6. Save configuration parameters and restart the motor.
01 10 00 54 00 02 04 00 01 00 01 67 60
7. Perform "return to zero"
01 06 00 35 00 00 99 C4

It can be observed that: the motor rotates forward at high speed (100RPM) → the switch is triggered → the motor rotates in reverse at low speed (10RPM) → the motor rotates forward at low speed (10RPM) → the switch is triggered again → the motor runs at high speed to the offset position and returns to zero.

```
[2025-10-27 09:51:10.246]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 09:51:10.303]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 09:51:15.815]# SEND HEX>
01 06 00 38 00 00 08 07
[2025-10-27 09:51:15.871]# RECV HEX>
01 06 00 38 00 00 08 07
[2025-10-27 09:51:20.918]# SEND HEX>
01 10 00 2A 00 04 08 00 00 00 00 00 64 00 0A EE 05
[2025-10-27 09:51:20.994]# RECV HEX>
01 10 00 2A 00 04 08 00 00 00 00 00 64 00 0A EE 05
[2025-10-27 09:51:26.806]# SEND HEX>
01 10 00 2E 00 02 04 20 00 00 00 7B FB
[2025-10-27 09:51:26.874]# RECV HEX>
01 10 00 2E 00 02 04 20 00 00 00 7B FB
[2025-10-27 09:51:32.351]# SEND HEX>
01 10 00 31 00 03 06 00 00 EA 60 00 00 83 BC
[2025-10-27 09:51:32.426]# RECV HEX>
01 10 00 31 00 03 06 00 00 EA 60 00 00 83 BC
[2025-10-27 09:51:38.207]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-27 09:51:38.269]# RECV HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-27 09:51:45.775]# SEND HEX>
01 06 00 35 00 00 99 C4
[2025-10-27 09:51:45.845]# RECV HEX>
01 06 00 35 00 00 99 C4
```



6.4.2 Example of automatic return to zero upon power-on

The setup steps are as follows:

1. Configure **working mode**
01 06 00 10 00 05 48 0C
2. Configure **the active level of the I/O input of the switch.**
01 06 00 38 00 00 08 07
3. Configure **return-to-zero mode, direction, "High Speed", "Low Speed"**
01 10 00 2A 00 04 08 00 00 00 00 00 64 00 0A EE 05
4. Configure **origin offset**
01 10 00 2E 00 02 04 20 00 00 00 7B FB
5. Configure the zero-return **trigger method** and **timeout parameters.**
01 10 00 31 00 03 06 00 01 EA 60 00 00 BE 7C
6. Save configuration parameters
01 06 00 54 00 01 09 DA

After a power outage and restart, the following can be observed: the motor rotates at high speed (100 RPM) in the forward direction → the switch is triggered → the motor rotates in the reverse direction at low speed (10 RPM) → the motor rotates in the forward direction at low speed (10 RPM) → the switch is triggered again → the motor runs at high speed to the offset position and returns to zero.

```
[2025-10-27 09:59:55.191]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 09:59:55.253]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 10:00:02.382]# SEND HEX>
01 06 00 38 00 00 08 07
[2025-10-27 10:00:02.452]# RECV HEX>
01 06 00 38 00 00 08 07
[2025-10-27 10:00:09.176]# SEND HEX>
01 10 00 2A 00 04 08 00 00 00 00 00 64 00 0A EE 05
[2025-10-27 10:00:09.251]# RECV HEX>
01 10 00 2A 00 04 E0 02
[2025-10-27 10:00:15.510]# SEND HEX>
01 10 00 2E 00 02 04 20 00 00 00 7B FB
[2025-10-27 10:00:15.570]# RECV HEX>
01 10 00 2E 00 02 21 C1
[2025-10-27 10:00:29.039]# SEND HEX>
01 10 00 31 00 03 06 00 01 EA 60 00 00 BE 7C
[2025-10-27 10:00:29.095]# RECV HEX>
01 10 00 31 00 03 D1 C7
[2025-10-27 10:00:50.863]# SEND HEX>
01 06 00 54 00 01 09 DA
[2025-10-27 10:00:50.927]# RECV HEX>
01 06 00 54 00 01 09 DA
```

Note: If the limit switch is not triggered within the set "return to zero timeout period", then "return to zero timeout failure" will occur.



6.5 Example of mechanical limit switch homing configuration

A suitable zero-return current needs to be set for mechanical limit switches. The zero-return current should not be too large, just enough to drive the load, to avoid damaging the mechanical device.

6.5.1 Example of mechanical limit command triggering return to zero

1. Configure **working mode**

01 06 00 10 00 05 48 0C

2. Configure **return-to-zero mode, direction, "High Speed", "Low Speed"**

01 10 00 2A 00 04 08 00 01 00 00 00 64 00 0A FE C5

3. Configure **the origin offset , Return-to-zero torque**

01 10 00 2E 00 03 06 20 00 00 00 02 58 81 E5

4. Configure the zero-return **trigger method** and **timeout parameters.**

01 10 00 31 00 03 06 00 00 EA 60 00 00 83 BC

5. Save configuration parameters and restart the motor.

01 10 00 54 00 02 04 00 01 00 01 67 60

6. Perform "return to zero"

01 06 00 35 00 00 99 C4

It can be observed that: the motor rotates forward at low speed (10RPM) → after touching the mechanical limit switch → the motor stops → the motor runs in reverse at high speed to the offset position, and the return to zero is completed.

```
[2025-10-27 10:18:05.392]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 10:18:05.455]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 10:18:19.409]# SEND HEX>
01 10 00 2A 00 04 08 00 01 00 00 00 64 00 0A FE C5
[2025-10-27 10:18:19.486]# RECV HEX>
01 10 00 2A 00 04 E0 02
[2025-10-27 10:18:27.896]# SEND HEX>
01 10 00 2E 00 03 06 20 00 00 00 02 58 81 E5
[2025-10-27 10:18:27.970]# RECV HEX>
01 10 00 2E 00 03 E0 01
[2025-10-27 10:18:44.680]# SEND HEX>
01 10 00 31 00 03 06 00 00 EA 60 00 00 83 BC
[2025-10-27 10:18:44.756]# RECV HEX>
01 10 00 31 00 03 D1 C7
[2025-10-27 10:18:52.976]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-27 10:18:53.045]# RECV HEX>
01 10 00 54 00 02 00 18
[2025-10-27 10:19:01.256]# SEND HEX>
01 06 00 35 00 00 99 C4
[2025-10-27 10:19:01.316]# RECV HEX>
01 06 00 35 00 00 99 C4
```

Note: If the motor does not touch the mechanical limit within the set "return to zero timeout period", then "return to zero timeout failure" will occur.



6.5.2 Example of automatic zeroing upon power-on of mechanical limit switch

1. Configure **working mode**
01 06 00 10 00 05 48 0C
2. Configure **return-to-zero mode**, **direction**, "**High Speed**", "**Low Speed**"
01 10 00 2A 00 04 08 00 01 00 00 00 64 00 0A FE C5
3. Configure **the origin offset** , **Return-to-zero torque**
01 10 00 2E 00 03 06 20 00 00 00 02 58 81 E5
4. Configure the **zero-return trigger method** and **timeout parameters**.
01 10 00 31 00 03 06 00 01 EA 60 00 00 BE 7C
5. **Save configuration parameters**
01 06 00 54 00 01 09 DA

After a power outage and restart, the following can be observed: the motor rotates forward at low speed (10 RPM) → after touching the mechanical limit switch → the motor stops → the motor runs in reverse at high speed to the offset position, and the return to zero is completed.

```
[2025-10-27 14:05:16.493]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 14:05:16.559]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 14:05:22.725]# SEND HEX>
01 10 00 2A 00 04 08 00 01 00 00 00 64 00 0A FE C5
[2025-10-27 14:05:22.805]# RECV HEX>
01 10 00 2A 00 04 E0 02
[2025-10-27 14:05:31.341]# SEND HEX>
01 10 00 2E 00 03 06 20 00 00 02 58 81 E5
[2025-10-27 14:05:31.413]# RECV HEX>
01 10 00 2E 00 03 E0 01
[2025-10-27 14:05:41.238]# SEND HEX>
01 10 00 31 00 03 06 00 01 EA 60 00 00 BE 7C
[2025-10-27 14:05:41.303]# RECV HEX>
01 10 00 31 00 03 D1 C7
[2025-10-27 14:05:55.893]# SEND HEX>
01 06 00 54 00 01 09 DA
[2025-10-27 14:05:55.959]# RECV HEX>
01 06 00 54 00 01 09 DA
```

Note: If the motor does not touch the mechanical limit within the set "return to zero timeout period", then "return to zero timeout failure" will occur.



6.6 Example of Single-lap zero-return configuration

To return to zero in a single lap, the "coordinate zero point" within the single lap must be set in advance using command 92H.

6.6.1 Example of single-lap command triggering zero return

First, move the motor shaft to the appropriate "zero point" position.

1. Configure **working mode**

```
01 06 00 10 00 05 48 0C
```

2. Configure return-to-zero **mode**, selecting the nearest option, "**High Speed**", "**Low Speed**"

```
01 10 00 2A 00 04 08 00 02 00 02 00 64 00 0A CD C5
```

3. Configure the zero-return **trigger method** and **timeout parameters**.

```
01 10 00 31 00 03 06 00 00 EA 60 00 00 83 BC
```

4. Set the "**zero point**" coordinates

```
01 06 00 34 00 01 09 C4
```

5. Save configuration parameters and restart the motor.

```
01 10 00 54 00 02 04 00 01 00 01 67 60
```

6. Perform "return to zero"

```
01 06 00 35 00 00 99 C4
```

After saving the settings, you can move the motor shaft away from the "zero" position by turning off the power or loosening the shaft.

After executing "return to zero", it can be observed that the motor runs at high speed (100RPM) to the "zero point" position, and the return to zero is completed.

```
[2025-10-27 15:47:24.960]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 15:47:25.027]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 15:47:30.816]# SEND HEX>
01 10 00 2A 00 04 08 00 02 00 02 00 64 00 0A B4 05
[2025-10-27 15:47:30.891]# RECV HEX>
01 10 00 2A 00 04 E0 02
[2025-10-27 15:47:37.016]# SEND HEX>
01 10 00 31 00 03 06 00 00 EA 60 00 00 83 BC
[2025-10-27 15:47:37.079]# RECV HEX>
01 10 00 31 00 03 D1 C7
[2025-10-27 15:47:43.561]# SEND HEX>
01 06 00 34 00 01 09 C4
[2025-10-27 15:47:43.638]# RECV HEX>
01 06 00 34 00 01 09 C4
[2025-10-27 15:47:49.176]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-27 15:47:49.242]# RECV HEX>
01 10 00 54 00 02 00 18
[2025-10-27 15:48:11.288]# SEND HEX>
01 06 00 35 00 00 99 C4
[2025-10-27 15:48:11.359]# RECV HEX>
01 06 00 35 00 00 99 C4
```



6.6.2 Example of automatic zero return upon power-on in a single revolution

First, move the motor shaft to the appropriate "zero point" position.

1. Configure **working mode**

01 06 00 10 00 05 48 0C

2. Configure the return-to-zero **mode**, **selecting the nearest option**,
" **High Speed** " or " **Low Speed** "

01 10 00 2A 00 04 08 00 02 00 02 00 64 00 0A CD C5

3. Configure the zero-return **trigger method** and **timeout parameters**.

01 10 00 31 00 03 06 00 01 EA 60 00 00 83 BC

4. Set the "zero point" coordinates

01 06 00 34 00 01 09 C4

5. Save configuration parameters

01 06 00 54 00 01 09 DA

After power is cut off, move the motor shaft away from the "zero point" position.

After powering on again, the motor will runs at high speed (100RPM) to the "zero point" position and returns to zero.

```
[2025-10-27 15:50:38.073]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 15:50:38.134]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 15:50:44.024]# SEND HEX>
01 10 00 2A 00 04 08 00 02 00 02 00 64 00 0A B4 05
[2025-10-27 15:50:44.093]# RECV HEX>
01 10 00 2A 00 04 E0 02
[2025-10-27 15:50:53.176]# SEND HEX>
01 10 00 31 00 03 06 00 01 EA 60 00 00 BE 7C
[2025-10-27 15:50:53.247]# RECV HEX>
01 10 00 31 00 03 D1 C7
[2025-10-27 15:51:00.728]# SEND HEX>
01 06 00 34 00 01 09 C4
[2025-10-27 15:51:00.798]# RECV HEX>
01 06 00 34 00 01 09 C4
[2025-10-27 15:51:07.944]# SEND HEX>
01 06 00 54 00 01 09 DA
[2025-10-27 15:51:08.012]# RECV HEX>
01 06 00 54 00 01 09 DA
```



6.7 Example of coordinate zeroing

Before performing "coordinate zeroing", the coordinates of the "zero point" must be determined:

For switch-based or mechanical limit switch-based zeroing, "origin zeroing" must be performed first.

For the single-loop zero-return method, simply set the "zero point" coordinates.

1. Execute "Coordinate Zeroing"

01 06 00 35 00 0 1 58 04

Once the "zero point" coordinates are determined, regardless of the motor's position, executing "coordinate return to zero" will cause the motor to return to the zero point position at high speed.

Note: The motor must be enabled before executing "coordinate return to zero".

```
[2025-10-27 15:53:18.752]# SEND HEX>
01 06 00 35 00 01 58 04
[2025-10-27 15:53:18.823]# RECV HEX>
01 06 00 35 00 01 58 04 |
```



Part 7. Left and right limit switch instructions

The default limit switch connection method is shown in the table below:

	Left limit port	Right limit port	Remark
SERV057ES	IN_COM / IN1	IN_COM / IN2	
SERV042ES	IN+ / IN-	EN+/EN-	right limit mapping

7.1 Set limit switch parameters

Address: 37H-39H

Note: Set the zero-origin offset and torque parameters.

Data type: uint16_t

Data Explanation:

EndLimit limit enable

00 : Limit switch function disabled (default value)

01 : Enable limit function

FF: Keep the original settings unchanged.

Effective level when the End Level switch is closed

00 : Low level when the switch is closed (default value)

01 : High level when the switch is closed

FF: Keep the original settings unchanged.

reMap finite bit mapping switch

00: Disable right limit port mapping (default value, only valid for 42ES)

01: Enable right limit port mapping

FF: Keep the original settings unchanged.

The query frame is as follows:

Downlink message (host computer → SERV0xxES)														
addr	code	Start address		register quantity		bytes	37H		38H		39H		CRC16	
		H	L	H	L		H	L	H	L	H	L	H	L
01H	10H	00H	37H	00H	03H	06H	EndLimit		EndLevel		reMap			

If the write is successful, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	37H	00H	03H	31H	C6H

Limit enable description:

1. When EndLimit = 1, in bus control mode

Left limit switch triggered, motor stops moving to the left.

Right limit switch triggered, motor stops moving to the right.

2. After using the limit function for the first time or changing the limit parameters, you must perform a "return to zero" operation once.

3. The limit function is invalid in pulse control mode.



7.2 Example of Limit switch configuration

Connect the limit switch to the corresponding port.

1. Configure the operating mode (skip this step if bus mode is already configured).
01 06 00 10 00 05 48 0C
2. Set limit enable , trigger level , and right limit mapping.
57ES 01 10 00 37 00 03 06 00 01 00 00 00 00 6A 65 (No right limit mapping required)
42ES 01 10 00 37 00 03 06 00 01 00 00 00 01 AB A5 (Right limit mapping required)
3. Configure return-to-zero mode, direction, "High Speed", "Low Speed"
01 10 00 2A 00 04 08 00 00 00 00 00 64 00 0A EE 05
4. Save configuration parameters and restart the motor.
01 10 00 54 00 02 04 00 01 00 01 67 60
5. Perform "return to zero"
01 06 00 35 00 00 99 C4

(After using the limit function for the first time or changing the limit switch parameters, you must perform a "zeroing" operation once .)

```
[2025-10-28 11:36:10.127]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 11:36:10.196]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 11:36:16.070]# SEND HEX>
01 10 00 37 00 03 06 00 01 00 00 00 00 6A 65
[2025-10-28 11:36:16.144]# RECV HEX>
01 10 00 37 00 03 31 C6
[2025-10-28 11:36:22.390]# SEND HEX>
01 10 00 2A 00 04 08 00 00 00 00 00 64 00 0A EE 05
[2025-10-28 11:36:22.453]# RECV HEX>
01 10 00 2A 00 04 E0 02
[2025-10-28 11:36:29.070]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-28 11:36:29.144]# RECV HEX>
01 10 00 54 00 02 00 18
[2025-10-28 11:36:37.750]# SEND HEX>
01 06 00 35 00 00 99 C4
[2025-10-28 11:36:37.816]# RECV HEX>
01 06 00 35 00 00 99 C4
```



7.3 Example of Left limit switch test

Connect the limit switch to the corresponding port.

1. Configure running **speed** and **acceleration** parameters
01 10 00 41 00 02 04 01 2C 00 02 77 A7
2. Configure **the pulse count** and **run it relative to the target**.
01 10 00 48 00 03 06 E2 00 40 00 01 A6 3C

When the motor is running, triggering the left limit switch will stop the motor.

```
[2025-10-28 11:40:37.070]# SEND HEX>
01 10 00 41 00 02 04 01 2C 00 02 77 A7
[2025-10-28 11:40:37.136]# RECV HEX>
01 10 00 41 00 02 11 DC
[2025-10-28 11:40:47.567]# SEND HEX>
01 10 00 48 00 03 06 E2 00 40 00 01 A6 3C
[2025-10-28 11:40:47.638]# RECV HEX>
01 10 00 48 00 03 00 1E |
```

7.4 Example of Right limit switch test

Connect the limit switch to the corresponding port.

1. Configure running **speed** and **acceleration** parameters
01 10 00 41 00 02 04 01 2C 00 02 77 A7
2. Configure **the pulse count** and **run it relative to the target**.
01 10 00 48 00 03 06 E2 00 FC 00 01 83 AC

When the motor is running, triggering the right limit switch will stop the motor.

```
[2025-10-28 11:42:21.582]# SEND HEX>
01 10 00 41 00 02 04 01 2C 00 02 77 A7
[2025-10-28 11:42:21.662]# RECV HEX>
01 10 00 41 00 02 11 DC
[2025-10-28 11:42:27.846]# SEND HEX>
01 10 00 48 00 03 06 E2 00 FC 00 01 83 AC
[2025-10-28 11:42:27.907]# RECV HEX>
01 10 00 48 00 03 00 1E
```



Part 8. Bus control mode parameter description

Note 1 : The instructions in this section are only valid in bus control mode.

8.1 Curve acceleration/deceleration parameter description

the unit of rotational speed (RPM) : It is an abbreviation for Revolutions Per Minute, which means revolutions per minute.

Curve acceleration/deceleration control of motor operation involves two parameters: speed and acceleration (acceleration). These are explained below :

1.speed

The speed parameter ranges from 0 to 3000. The larger the value, the faster the motor rotates.

When speed = 0, the motor stops rotating.

If the set speed is greater than the maximum speed of the control mode, the motor runs at the maximum speed of the control mode.

Note: The speed value is calibrated based on 16/32/64 subdivisions, and the speeds of other subdivisions need to be calculated based on 16 subdivisions.

For example, setting speed=1200

At 8 subdivisions, the speed is 2400 (RPM)

At 16/32/64 subdivisions, the speed is 1200 (RPM)

At 128 subdivisions, the speed is 150 (RPM)



2. acceleration

The value of the acceleration(acc) ranges from 0 to 255. The larger the value, the faster the motor accelerates/decelerates.

If acc=0, the motor runs without acceleration or deceleration, and runs directly at the set speed.

① accelerates

Suppose at time t_1 , the current speed is V_{t1} ($V_{t1} < \text{speed}$)

at time t_2 , the current speed is V_{t2}

$$t_2 - t_1 = (256 - \text{acc}) * 50 (\mu\text{S})$$

The relationship between the current speed V_{ti} , acc, and speed is as follows:

$$V_{t2} = V_{t1} + 1 \quad (V_{t2} \leq \text{speed})$$

For example: acc = 236, speed = 3000

T(ms)	speed (RPM)
0	0
1	1
2	2
3	3
...	...

T(ms)	speed (RPM)
...	...
...	...
2998	2998
2999	2999
3000	3000

② decelerates

Suppose at time t_1 , the current speed is V_{t1} ($V_{t1} > \text{speed}$)

at time t_2 , the current speed is V_{t2}

$$t_2 - t_1 = (256 - \text{acc}) * 50 (\mu\text{S})$$

The relationship between the current speed V_{ti} , acc, and speed is as follows:

$$V_{t2} = V_{t1} - 1 \quad (V_{t2} \geq \text{speed})$$



8.2 Bus control general instructions

8.2.1 Read motor operating status

Address: 3CH

Data type: uint16_t

Note: Obtain the stall status of the motor: 00 Not stalled / 01 Stalled

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Numbers		CRC16	
		H	L	H	L	H	L
01H	03H	00H	3CH	00H	01H	44H	06H

If the motor stops, the response frame will be as follows:

Uplink message (host computer ← SERV0xxES)						
Address	code	Number of bytes	Register data		CRC16	
			H	L	H	L
01H	03H	02H	00H	01H	79H	84H

Motor status	status	Remark
Query failed	0	
Stop running	1	
Speed up	2	
slow down	3	
Run at full speed	4	
Reset to zero	5	



8.2.2 Set motor enable state

Address: 3DH

Note: Motor enable ON or OFF

Data type: uint16_t

Data range: 00 Enable OFF

01 Enable ON (default value)

Assuming the configuration is: motor enabled ON, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	3DH	00H	01H	D9H	C6H

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	3DH	00H	01H	D9H	C6H

Note: Although the data was written successfully, the actual configuration failed because it was out of range.

8.2.3 Emergency stop command for motor

Address: 3EH

Note: Emergency stop of motor

Data type: uint16_t

The query frame is as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	3EH	00H	01H	29H	C6H

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	3EH	00H	01H	29H	C6H



Part 9. Speed Control Mode Description

Speed control mode

It can control the motor to run continuously at a set acceleration and speed.

It can control the motor to run at a set acceleration and speed for a set period of time.

It allows the motor to run at a set speed and acceleration as soon as it is turned on.

9.1 Configure running speed , acceleration, and direction

Address: 41H-43H

Note: Configure running speed, acceleration, and direction.

Data type: uint16_t

Data Explanation:

speed value range 0-3000

acc values range from 0 to 255.

dir Direction, with a value range of 0 / 1 (C CW/CW)

The query frame is as follows:

Downlink message (host computer → SERV0xxES)														
addr	code	Start address		register quantity		bytes	31H		32H		33H		CRC16	
		H	L	H	L		H	L	H	L	H	L	H	L
01H	10H	00H	41H	00H	03H	06H	speed		acc		dir			

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	41H	00H	03H	D0H	1CH



9.2 Configure runtime

Address: 44H-45H

Note: Configure the running time of speed mode. Speed mode will automatically stop after the configured running time.

When the runtime is configured to 0, speed mode will run continuously and will not stop automatically.

Data type: uint32_t

Data unit: 10ms

Data Explanation: Runtime duration (default value 0, meaning no stopping)

Downlink message (host computer → SERV0xxES)												
addr	code	Start address		register quantity		bytes	44H		45H		CRC16	
		H	L	H	L		H	L	H	L	H	L
01H	10H	00H	44H	00H	02H	04H	runtime					

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	44H	00H	02H	01H	DDH

9.3 Run/Stop Commands

Address: 46H

Note: During operation, the motor runs according to the configured speed, acceleration, direction, and time.

When stopping, the motor stops at the configured acceleration.

Data type: uint16_t

Data range: 00 Stop

01 Running

Assuming the configuration is as follows: the motor is running, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	46H	00H	01H	A9H	DFH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	46H	00H	01H	A9H	DFH



9.4 Configure automatic operation

Address: 40H

Note: After power-on, the motor will run automatically according to the configured parameters.

This information must be kept valid.

Data type: uint16_t

Data range: 00 Disabled automatic operation

01 Enable automatic operation

Assuming the configuration is set to automatic operation, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	40H	00H	01H	49H	DEH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	40H	00H	01H	49H	DEH

Note: The save command (54H) must be executed, and it will be effective after power-on.



9.5 Example of Speed mode configuration

1. Configure the operating mode
01 06 00 10 00 05 48 0C
(skip this step if bus mode is already configured).
2. Save the configuration parameters and restart the motor
01 10 00 54 00 02 04 00 01 00 01 67 60
(skip this step if bus mode is already configured).
3. Configure running speed , acceleration , and direction parameters.
01 10 00 41 00 03 06 01 2C 00 02 00 00 84 46
4. Configure runtime parameters
01 10 00 44 00 02 04 27 10 00 00 FD 1D (100s)
(skip this step if automatic shutdown is not required).
5. Run command
01 06 00 46 00 01 A9 DF
The motor will running according to the configure parameters.
6. to slow down and stop the machine during operation , the command is as follows:
01 06 00 46 00 00 68 1F
The motor will stop at the configured acceleration.
7. If an emergency stop is required during operation , the command is as follows:
01 06 00 3E 00 01 29 C6

```
[2025-10-27 18:12:03.236]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 18:12:03.303]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 18:12:10.900]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-27 18:12:10.963]# RECV HEX>
01 10 00 54 00 02 00 18
[2025-10-27 18:12:21.243]# SEND HEX>
01 10 00 41 00 03 06 01 2C 00 02 00 00 84 46
[2025-10-27 18:12:21.309]# RECV HEX>
01 10 00 41 00 03 D0 1C
[2025-10-27 18:13:26.388]# SEND HEX>
01 10 00 44 00 02 04 27 10 00 00 FD 1D
[2025-10-27 18:13:26.448]# RECV HEX>
01 10 00 44 00 02 01 DD
[2025-10-27 18:13:39.004]# SEND HEX>
01 06 00 46 00 01 A9 DF
[2025-10-27 18:13:39.078]# RECV HEX>
01 06 00 46 00 01 A9 DF
[2025-10-27 18:13:47.267]# SEND HEX>
01 06 00 46 00 00 68 1F
[2025-10-27 18:13:47.344]# RECV HEX>
01 06 00 46 00 00 68 1F
```



9.6 Example of automatic running configuration

1. Configure working mode
01 06 00 10 00 05 48 0C
2. Configure running speed, acceleration, direction parameters.
01 10 00 41 00 03 06 01 2C 00 02 00 00 84 46
3. Configure automatic operation
01 06 00 40 00 01 49 DE
4. Save configuration parameters and restart the motor.
01 10 00 54 00 02 04 00 01 00 01 67 60

the motor will running with the configured parameters.

```
[2025-10-27 18:19:00.964]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 18:19:01.030]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-27 18:19:06.732]# SEND HEX>
01 10 00 41 00 03 06 01 2C 00 02 00 00 84 46
[2025-10-27 18:19:06.810]# RECV HEX>
01 10 00 41 00 03 06 01 2C 00 02 00 00 84 46
[2025-10-27 18:19:12.756]# SEND HEX>
01 06 00 40 00 01 49 DE
[2025-10-27 18:19:12.822]# RECV HEX>
01 06 00 40 00 01 49 DE
[2025-10-27 18:19:36.588]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-27 18:19:36.659]# RECV HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
```



Part 10. Position control mode description

10.1 Movement based on pulse count

In this mode, the motor can be controlled to run to a specified position relative to or absolutely based on the set acceleration and speed, according to the number of pulses.

10.1.1 Configure running speed and acceleration

Address: 41H-42H

Note: Configure running speed and acceleration.

Data type: uint16_t

Data Explanation:

speed value range 0-3000

acc values range from 0 to 255.

The query frame is as follows:

Downlink message (host computer → SERVO _{xx} ES)												
addr	code	Start address		register quantity		bytes	41H		42H		CRC16	
		H	L	H	L		H	L	H	L	H	L
01H	10H	00H	41H	00H	02H	04H	speed		acc			

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	41H	00H	02H	11H	DCH

10.1.2 Configure pulse count

Address: 48H-49H

Note: Configure pulse count

Data type: int32_t

Assuming the configuration is: pulse count - 32000, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)												
addr	code	Start address		register quantity		bytes	48H		49H		CRC16	
		H	L	H	L		H	L	H	L	H	L
01H	10H	00H	48H	00H	02H	04H	83H	00H	FFH	FFH	DEH	0DH

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	48H	00H	02H	C1H	DEH



10.1.3 Run/Stop according to pulse command

Address: 4AH

Note: During operation, the motor runs at the configured speed, acceleration, and pulse count.

When stopped, the motor stops at the configured acceleration.

Data type: uint16_t

Data Explanation:

- H 00 relative pulse position
- 01 Absolute Pulse Position
- L 00 Stop
- 01 Running

Assuming the configuration is relative to the pulse position, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	4AH	00H	01H	A8H	1CH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	4AH	00H	00H	A8H	1CH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



10.1.4 Example of Pulse Number Relative Running

1. Configure the operating mode (**skip this step if bus mode is already configured**).
01 06 00 10 00 05 48 0C
2. Save the configuration parameters and restart the motor (**skip this step if bus mode is already configured**) .
01 10 00 54 00 02 04 00 01 00 01 67 60
3. Configure running **speed** and **acceleration** parameters
01 10 00 41 00 02 04 01 2C 00 02 77 A7
4. Configure **the pulse count** and **run it relative to the target**.
01 10 00 48 00 03 06 E2 00 00 04 00 01 F2 3D
the motor will running according to the configured parameters.
5. **to slow down and stop the machine** during operation , the command is as follows:
01 06 00 4A 00 00 A8 1C
the motor will stops at the configured acceleration.
6. **If an emergency stop** is required during operation , the command is as follows:
01 06 00 3E 00 01 29 C6

```
[2025-10-28 09:30:01.715]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 09:30:01.786]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 09:30:09.011]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-28 09:30:09.075]# RECV HEX>
01 10 00 54 00 02 00 18
[2025-10-28 09:30:16.875]# SEND HEX>
01 10 00 41 00 02 04 01 2C 00 02 77 A7
[2025-10-28 09:30:16.946]# RECV HEX>
01 10 00 41 00 02 11 DC
[2025-10-28 09:30:25.155]# SEND HEX>
01 10 00 48 00 03 06 E2 00 00 04 00 01 F2 3D
[2025-10-28 09:30:25.218]# RECV HEX>
01 10 00 48 00 03 00 1E
[2025-10-28 09:30:42.507]# SEND HEX>
01 06 00 4A 00 00 A8 1C
[2025-10-28 09:30:42.573]# RECV HEX>
01 06 00 4A 00 00 A8 1C
```



10.1.5 Example of Pulse Count Absolute Running

1. Configure the operating mode (**skip this step if bus mode is already configured**).
01 06 00 10 00 05 48 0C
2. Save the configuration parameters and restart the motor (**skip this step if bus mode is already configured**) .
01 10 00 54 00 02 04 00 01 00 01 67 60
3. Configure the "zero point" (**skip this step if the system has already returned to zero or if zero point configuration is not required**).
01 06 00 34 00 01 09 C4
4. Configure running **speed** and **acceleration** parameters
01 10 00 41 00 02 04 01 2C 00 02 77 A7
5. Configure **the pulse count** and **run in absolute mode**.
01 10 00 48 00 03 06 E2 00 00 04 01 01 F3 AD
the motor will running according to the configured parameters.
6. **to slow down and stop the machine** during operation , the command is as follows:
01 06 00 4A 01 00 A9 8C
The motor will stops at the configured acceleration.
7. **If an emergency stop** is required during operation , the command is as follows:
01 06 00 3E 00 01 29 C6

```
[2025-10-28 10:10:29.860]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 10:10:29.931]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 10:10:37.788]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-28 10:10:37.855]# RECV HEX>
01 10 00 54 00 02 00 18
[2025-10-28 10:10:44.716]# SEND HEX>
01 06 00 34 00 01 09 C4
[2025-10-28 10:10:44.776]# RECV HEX>
01 06 00 34 00 01 09 C4
[2025-10-28 10:10:51.108]# SEND HEX>
01 10 00 41 00 02 04 01 2C 00 02 77 A7
[2025-10-28 10:10:51.163]# RECV HEX>
01 10 00 41 00 02 11 DC
[2025-10-28 10:10:57.324]# SEND HEX>
01 10 00 48 00 03 06 E2 00 00 04 01 01 F3 AD
[2025-10-28 10:10:57.381]# RECV HEX>
01 10 00 48 00 03 00 1E
[2025-10-28 10:11:04.268]# SEND HEX>
01 06 00 4A 01 00 A9 8C
[2025-10-28 10:11:04.344]# RECV HEX>
01 06 00 4A 01 00 A9 8C
```



10.2 Move according to coordinate values

In this mode, the motor can be controlled to move to the specified position according to the coordinate values at the set acceleration and speed.

10.2.1 Configure running speed and acceleration

Address: 41H-42H

Note: Configure running speed and acceleration.

Data type: uint16_t

Data Explanation:

speed value range 0-3000
acc values range from 0 to 255.

The query frame is as follows:

Downlink message (host computer → SERVO _{xx} ES)												
addr	code	Start address		register quantity		bytes	41H		42H		CRC16	
		H	L	H	L		H	L	H	L	H	L
01H	10H	00H	41H	00H	02H	04H	speed		acc			

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	41H	00H	02H	11H	DCH

10.2.2 Configure coordinate values

Address: 4CH-4DH

Note: Configure coordinate values

Data type: int32_t

Assuming the configuration is: coordinates **-262144**, the query frame would look like this:

Downlink message (host computer → SERV0xxES)												
addr	code	Start address		register quantity		bytes	48H		49H		CRC16	
		H	L	H	L		H	L	H	L	H	L
01H	10H	00H	4CH	00H	02H	04H	00H	00H	FFH	FCH	B6H	7BH

The response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Numbers written		CRC16	
		H	L	H	L	H	L
01H	10H	00H	4CH	00H	02H	80H	1FH



10.2.3 Run/Stop Commands by Coordinates

Address: 4EH

Note: During operation, the motor runs according to the configured speed, acceleration, and coordinate values.

When stopped, the motor stops at the configured acceleration.

Data type: uint16_t

Data Explanation:

- H 00 Relative coordinate position
- 01 Absolute coordinate position
- L 00 Stop
- 01 Running

Assuming the configuration is relative to the coordinate position, the query frame will be as follows:

Downlink message (host computer → SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	4EH	00H	01H	28H	1DH

After successful writing, the response frame is as follows:

Uplink message (host computer ← SERV0xxES)							
Address	code	Start address		Data		CRC16	
		H	L	H	L	H	L
01H	06H	00H	4EH	00H	01H	28H	1DH

Note: Although the data was written successfully, the actual configuration failed because it was out of range.



10.2.4 Example of Relative coordinate values running

1. Configure the operating mode (**skip this step if bus mode is already configured**).
01 06 00 10 00 05 48 0C
2. Save the configuration parameters and restart the motor (**skip this step if bus mode is already configured**) .
01 10 00 54 00 02 04 00 01 00 01 67 60
3. Configure running **speed** and **acceleration** parameters
01 10 00 41 00 02 04 01 2C 00 02 77 A7
4. Configure **coordinate values** and **run relative to them**.
01 10 00 4C 00 03 06 00 00 00 19 00 01 34 6C
the motor will running according to the configured parameters.
5. **to slow down and stop the machine** during operation , the command is as follows:
01 06 00 4E 00 00 E9 DD
the motor will stops at the configured acceleration.
6. **If an emergency stop** is required during operation , the command is as follows:
01 06 00 3E 00 01 29 C6

```
[2025-10-28 10:59:56.461]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 10:59:56.525]# RCV HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 11:00:02.861]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-28 11:00:02.924]# RCV HEX>
01 10 00 54 00 02 00 18
[2025-10-28 11:00:08.693]# SEND HEX>
01 10 00 41 00 02 04 01 2C 00 02 77 A7
[2025-10-28 11:00:08.755]# RCV HEX>
01 10 00 41 00 02 11 DC
[2025-10-28 11:00:22.413]# SEND HEX>
01 10 00 4C 00 03 06 00 00 00 19 00 01 34 6C
[2025-10-28 11:00:22.485]# RCV HEX>
01 10 00 4C 00 03 41 DF
[2025-10-28 11:00:27.877]# SEND HEX>
01 06 00 4E 00 00 E9 DD
[2025-10-28 11:00:27.937]# RCV HEX>
01 06 00 4E 00 00 E9 DD
```



10.2.5 Example of absolute coordinate values Running

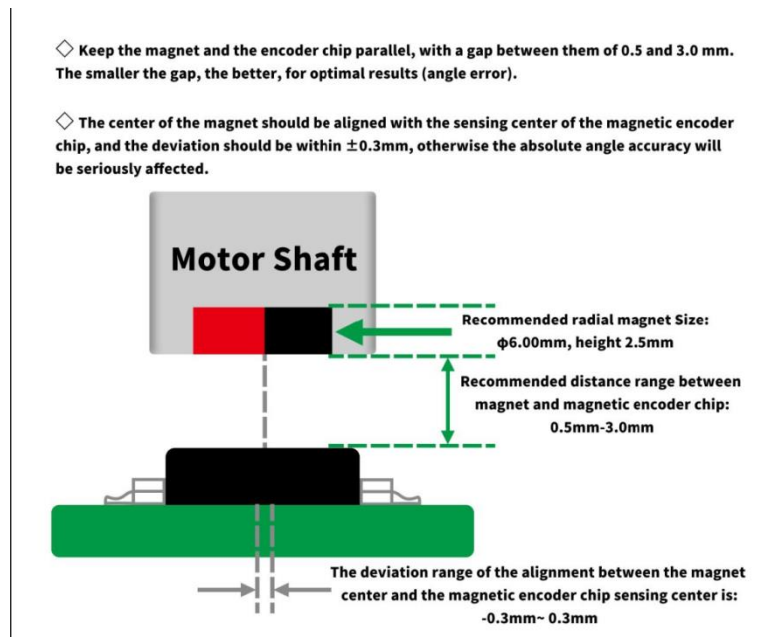
1. Configure the operating mode (**skip this step if bus mode is already configured**).
01 06 00 10 00 05 48 0C
2. Save the configuration parameters and restart the motor (**skip this step if bus mode is already configured**) .
01 10 00 54 00 02 04 00 01 00 01 67 60
3. Configure the "zero point" (**skip this step if the system has already returned to zero or if zero point configuration is not required**).
01 06 00 34 00 01 09 C4
4. Configure running **speed** and **acceleration** parameters
01 10 00 41 00 02 04 01 2C 00 02 77 A7
5. Configure **the pulse count** and **run in absolute mode**.
01 10 00 4C 00 03 06 00 00 00 19 01 01 35 FC
the motor will running according to the configured parameters.
6. **to slow down and stop the machine** during operation , the command is as follows:
01 06 00 4E 01 00 E8 4D
the motor will stops at the configured acceleration.
7. **an emergency stop** is required during operation , the command is as follows:
01 06 00 3E 00 01 29 C6

```
[2025-10-28 11:15:53.525]# SEND HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 11:15:53.602]# RECV HEX>
01 06 00 10 00 05 48 0C
[2025-10-28 11:15:59.118]# SEND HEX>
01 10 00 54 00 02 04 00 01 00 01 67 60
[2025-10-28 11:15:59.195]# RECV HEX>
01 10 00 54 00 02 00 18
[2025-10-28 11:16:13.062]# SEND HEX>
01 06 00 34 00 01 09 C4
[2025-10-28 11:16:13.125]# RECV HEX>
01 06 00 34 00 01 09 C4
[2025-10-28 11:16:19.510]# SEND HEX>
01 10 00 41 00 02 04 01 2C 00 02 77 A7
[2025-10-28 11:16:19.567]# RECV HEX>
01 10 00 41 00 02 11 DC
[2025-10-28 11:16:26.606]# SEND HEX>
01 10 00 4C 00 03 06 00 00 00 19 01 01 35 FC
[2025-10-28 11:16:26.668]# RECV HEX>
01 10 00 4C 00 03 41 DF
[2025-10-28 11:16:32.638]# SEND HEX>
01 06 00 4E 01 00 E8 4D
[2025-10-28 11:16:32.715]# RECV HEX>
01 06 00 4E 01 00 E8 4D
```

Part 11. Frequently Asked Questions and Precautions

11.1 Precautions

1. Power input voltage: 20V - 48V ;
2. Do not plug or unplug the power cord or signal cable while the circuit is powered on, as this may damage the driver board .
3. The A+, A-, B+, and B- phase lines must correspond; otherwise, an error will occur.
4. We recommend that users purchase our matching motors directly to avoid the risk of incompatibility. If you choose to use a custom motor in closed-loop mode instead of our motors, the following conditions must be met:
 - (1) The motor step distance is 1.8 degrees.
 - (2) The motor's internal resistance is less than 10 ohms.
 - (3) A radial magnet can be installed on the back of the motor.
 - (4) When installing magnets, please note the following diagram:



5. After the magnets are installed, a calibration procedure must be performed to compensate for any installation deviations of the magnets; refer to section 4.1.



11.2 Frequently Asked Questions

Serial Number	question	Solution
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
1 2		

Part 12. contact us

<https://makerbase.aliexpress.com/>

<https://www.youtube.com/channel/UC2i5I1tcOXRJ2ZJiRxwpCUQ>

<https://github.com/makerbase-motor>